

PCS-9567MV-10000
Medium Voltage Skid
Commissioning Manual

Preface

About This Manual

The manual describes the PCS-9567MV-10000 Medium Voltage (MV) Skid in terms of safety information, commissioning and etc. Please read all safety information and get familiar with the functions and features before proceeding with the operations of this product.

Intended Audience

This manual is intended for the persons who are required to install, configure or service equipment, or any other associated operation, it assumes a reasonable level of understanding in these disciplines.

Only qualified personnel are allowed to install, operate and maintain this equipment. Qualified personnel are individuals who:

- Are familiar with the installation, commissioning, and operation of the equipment and of the system to which it is being connected.
- Are able to safely perform switching operations in accordance with accepted safety engineering practices and are authorized to energize and de-energize equipment and to isolate, ground, and label it.
- Are trained in the care and use of safety apparatus in accordance with safety engineering practices.
- Are trained in emergency procedures (first aid).

Symbols

Safety information is highlighted with the following symbols according to the degree of danger:



Indicates an imminently hazardous situation that, if not avoided, will result in death or serious injury.



Indicates a potentially hazardous situation that, if not avoided, could result in death or serious injury.



Indicates a potentially hazardous situation that, if not avoided, may result in minor or moderate injury or equipment damage.

NOTICE

Indicates that property damage can result if the measures specified are not taken.

INFORMATION

Important information about the product, please pay attention to avoid undesired result.

Document Revision History

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R1.00	2025-03-14	First release

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The information in this manual is carefully checked periodically, and necessary corrections will be included in future editions. If nevertheless any errors are detected, suggestions for correction or improvement are greatly appreciated.

We reserve the rights to make technical improvements without notice.

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1 Safety Instruction

This chapter presents the safety instructions for PCS-9567MV-10000 medium voltage skid (abbreviated as “**MV skid**”, or “**this product**”). Read this document thoroughly before operations. NR will not be liable for any damage caused by violation of the safety instructions in this document.

1.1 Personnel Requirements

- Only certified and professionally authorized personnel are allowed to work on or near the MV skid. All installation and service personnel must be trained to be familiar with the precautions and workflows mentioned in this manual, as well as the safety regulations.
- Strictly abide by all instructions contained in this manual, as well as the rules and laws of the country/region where the project is located.
- The minimum required safety equipment is as follows: safety shoes, safety gloves, safety glasses, head protection, work clothes. Take special precautions when you work close to exposed conductors.
- Only personnel with corresponding qualifications are permitted for special operations. Necessary safety precautions (such as fastening safety belts for climbing operations) must be taken.
- In case of abnormal conditions (such as abnormal noise, sudden bad weather, etc.), the operation should be stopped.
- Keep all doors locked while this product is operating. Spend as little time as possible near this product.

1.2 Electrical Connection

- Power cables and communication cables should be placed in different cable trenches. Communication cables should be laid as far away as possible from the power cables, and avoid power cables and communication cables walking parallel to each other for a long distance, so as to reduce electromagnetic interference caused by output voltage transients.
- There is large current in the power cable, so the damage of the cable insulation should be avoided during the installation process, so as to avoid short circuit. Moreover, the power cable should be fixed properly, and the connected cable should be carefully handled to avoid exerting excessive external force.
- Do NOT touch exposed terminals when the MV skid is energized, so as to avoid possible dangerous high voltage.

1 Safety Instruction

- Dangerous voltages may still be present in the DC circuit after the power is turned off in this product. These voltages will not disappear within several seconds.
- When connecting the AC voltage/current circuit or DC power supply circuit, please confirm the consistency with the nominal parameters of this product.
- When this control device is energized, it is forbidden to insert or unplug the PCB (Printed Circuit Board), otherwise it may cause the device to operate incorrectly.
- When connecting the output contacts of this control device to external circuit, the external power supply voltage should be carefully checked, so as to prevent the connected circuit from overheating.
- The grounding cable must be grounded reliably! Otherwise, this product may fail to work, and even may cause fatal shock hazard to the operator in case of malfunction.

1.3 Daily Operations

DANGER

The DC voltage may reach 1500V and the AC voltage may reach several tens of kilovolts when this product is running, and it will be dangerous to touch the live part of this product directly. Improper operation may lead to serious personal injury.

WARNING

Do NOT modify parameters of the MV skid without the permission of NR. Unauthorized modification of functional parameters may cause personal injury or equipment damage, in which case NR will not provide warranty service.

WARNING

When the primary system is in live operation, secondary side open circuit of the current transformer (CT) connected to this product is absolutely forbidden. Open circuit may cause extremely dangerous high voltages.

1.4 Maintenance & Replacement

- All maintenance and repair operations can only be performed on the equipment when the MV skid is completely disconnected from the battery system and the power grid, and it is confirmed that these power supplies will not be turned on again and the capacitor discharging is completed.

- Energy storage battery side open-circuit voltage must NOT exceed the maximum permissible input voltage, otherwise this product may be damaged.
- Grid-connected charging and discharging of MV skid should be approved by the local power supply department, and the relevant operation must be performed by certified and professionally authorized personnel.
- During the normal operation of the MV skid, do NOT disconnect the AC side circuit breaker (abbreviated as CB) or the DC side switch directly at will, otherwise personal injury or equipment damage may occur!
- Do NOT cut off the power supply of the PCS (Power Conversion System) control unit during operation. If it is required to cut off the power, please follow the shutdown operation process.
- Do NOT let the PCS run in the overload situation for a long time, please pay attention to the temperatures of the PCS and power modules in case of overload. If the temperature of the power module is high, please reduce the P (active power) value of the PCS remote regulation, so as to ensure the safety of operation.
- During maintenance, please make sure that the MV skid will not be energized accidentally. It is recommended to take insulation protection measures when contacting possible live parts.
- Clean the air filters in time to ensure smooth air flow.

1.5 Environmental Requirements

- The installation environment is outdoor.
- Please make sure that the installation environment is free from shaking and vibration.
- The foundation must meet the design requirements, ensuring it is solid, safe and reliable. The foundation must have the bearing capacity to support the weight of the MV skid. No depressions or inclinations are allowed on the ground surface. To prevent the risk of corrosion, the MV skid should be installed at a location higher than the surrounding environment, so that surface water does not accumulate around it.
- Adequate distance must be maintained between the installation location and walls or other equipment to meet the requirements for the maintenance access, escape route and ventilation.
- The installation environment should be as far away as possible from areas where people lives, as the MV skid may generate some noise during operation.
- Ensure that all external cable entries are completely sealed to prevent the ingress of foreign objects such as rats, insects and dust.
- During storage and installation, avoid unnecessarily opening the doors to protect the equipment from rain and dust.



2 Product Introduction

2.1 Overview

2.1.1 Technical Data

Table 2-1 Technical data of this product

Transformer	
Rated Power (kVA)	10000 [•] / 9000 [°] / 8000 [°]
Transformer Model	Oil type ^[1]
Transformer Vector	Dy11-y11
Protection Level	IP54 [•] / IP55 [°]
Anti-corrosion Grade	C4-H ^{•[2]} / C4-VH [°] / C5-M [°] / C5-H [°] / C5-VH [°]
Cooling Method	ONAN [•] / ONAF [°]
Temperature Rise	60K (Top Oil) 65K (Winding) @40℃
Oil Retention Tank	None [•] / Galvanized steel ^{°[3]}
Winding Material	Aluminum [•] / Copper [°]
Transformer Oil	25#/45# mineral oil [•] / Natural ester insulation oil [°]
Transformer Efficiency	IEC standard [•] / IEC Tier-2 [°]
MV Operating Voltage Range (kV)	11~33±5%
Nominal Frequency (Hz)	50 / 60
Altitude (m)	<1000 [•] / >1000 [°]
Switchgear	
Switchgear Type	Ring Main Unit, CCV ^[4]
Rated voltage (kV)	12/24/36
Insulating medium	SF ₆
Rated frequency (Hz)	50/60
Enclosure protection degree	IP3X
Gas tank protection degree	IP67
Gas leakage rate per year	≤0.1%
Rated Operating Current (A)	630
Switchgear Short Circuit Rating (kA/s)	20kA/3s [•] / 25kA/3s [°]
Switchgear IAC (kA/s)	A FL 20kA 1s
PCS * 4	
DC Input Voltage Range (V)	1050~1500

2 Product Introduction

Max. DC input Current (A)	1310*2
DC Voltage Ripple	< 2%
DC Current Ripple	< 3%
LV Nominal Operating Voltage (V)	690
LV Operating Voltage Range (V)	621~759
PCS Efficiency	98.7% ^[5]
Max. AC Output Current (A)	1151*2
Total Harmonic Distortion Rate	< 3%
Reactive Power Compensation	Four quadrant operation
Nominal Output Power (kVA)	1250*2
Max. AC Power (kVA)	1375*2
Power Factor Range	>0.99
Nominal Frequency (Hz)	50 / 60 Hz
Operating Frequency (Hz)	45~55 / 55~65 Hz
Connection Phases	Three-phase, three-wire system
Protection	
DC Input Protection	Disconnecter + Fuse inside of inverter
AC Output Protection	Motorized circuit breaker inside of inverter
DC Overvoltage Protection	Surge arrester, type II [•] / I+II [°]
AC Overvoltage Protection	Surge arrester, type II [•] / I+II [°]
Ground Fault Protection	DC IMD [•] / DC IMD+ AC IMD [°]
Transformer Protection	Protection relay for pressure, temperature, gassing
Fire Extinguishing System	Smoke detector sensor (dry contact)
Communication Interface	
Communication Method	CAN / RS-485 / RJ45 / Optical fiber
Supported Protocol	CAN / Modbus / IEC60870-5-103 / IEC 61850
Ethernet Switch Quantity	One for standard ^[6]
UPS	1kVA for 15min [•] / 1h [°] / 2h [°]
Skid General	
Dimensions (W*H*D)(mm)	12192*2896*2438 (40ft)
Weight (kg)	38800
Protection Level	IP54
Operating Temperature (°C)	-35~60, >45 derating
Storage Temperature (°C)	-40~70
Maximum Altitude (above sea level) (m)	5000, ≥3000 derating ^[7]
Environment Humidity	0~ 100%, No condensation
Type of Ventilation	Nature air cooling [•] / Forced air cooling [°]
Auxiliary Power Consumption (kVA)	21 (peak)
Auxiliary Transformer (kVA)	Without [•] / With [°] ^[8]

INFORMATION

- Standard ○ Optional

[1] If dry transformer is required, please contact with NR for more information.

[2] Lower protection level will be covered by C4-H.

[3] Standard for no supply of oil retention tank. If required to be integrated with PCS skid, please contact with NR.

[4] If other type of switchgear is required, please contact with NR for more information. In “CCV”, “C” is the abbreviation for “**Cable load break switch unit**”, “V” is the abbreviation for “**Vacuum circuit breaker unit**”.

[5] Typical discharge value for each PCS running at DC 1200V under IEC 62933-2-1 environment condition.

[6] If more Ethernet switch is required, please contact with NR for more information.

[7] When altitude is between 3000~4000m, the system LV AC voltage shall be less than 600V; When altitude is between 4000~4500m, the system LV AC voltage shall be less than 550V; When altitude is between 4500~6000m, please contact with NR for more information.

[8] Please contact with NR for more information.

2.1.2 Functions

- **Accurate and flexible charging & discharging control modes**

Provides CAN, Ethernet and RS-485 interface for real-time communication with battery management system (BMS). It can accurately monitor current battery operation information, not only control the charging and discharging status of converter, but also switch to constant current mode, constant voltage mode, constant power mode and other charging & discharging modes conveniently, so as to achieve the optimal strategy for the battery charge and discharge control. Various types of energy storage batteries are supported: lithium-ion battery, sodium-ion battery, lead-acid battery, lead-carbon battery and etc.

- **Free switchover between grid-connected operation mode and grid-forming operation mode**

Realizes bi-directional energy exchange with the power grid in grid-connected mode, also can support regional power grid in grid-forming mode. Moreover, it can be used as the main power supply to support the isolated grid. The 2 operation modes can be switched freely.

- **Soft grid-connected control and power quality control**

The control system provides accurate real-time control for the output voltage of the converter

according to the monitored online power grid voltage. It can eliminate the static & dynamic errors, and realize the impact-free grid-connection. The optimized filter and harmonic suppression algorithm can realize the optimal control of grid-connected power and ensure the power quality.

- **Respond to EMS instructions for load leveling**

Under the dispatching of EMS (Energy Management System), it can store electric energy during load valley and release electric energy during load peak, so as to reduce peak-valley difference of power grid, improve load characteristics of power grid, and realize power system load control and load shifting.

- **Power grid frequency and reactive power regulation**

During grid-connected operation, it provides primary and secondary frequency regulation in coordination with AGC (Automatic Generation Control) or PMS (Power Management System). Moreover, it can realize steady-state reactive power control in coordination with AVC (Automatic Voltage Control), realize transient voltage control in coordination with PMS.

- **Complete self-supervision and protection functions**

The scope of self-supervision covers the control system, I/O units, converter power modules, buses and other related equipment. The system fault can be detected within 1ms and corresponding blocking pulse or tripping signal will be issued if fault is detected. The complete protections ensure the normal operation of the converter.

- **Powerful communication function**

The device is equipped with multiple optical interfaces and RJ45 interfaces, and supports high-speed communication protocols (e.g.: GOOSE and IEC 60044-8). The communication time delay is less than 2ms, and it can meet the fast control response requirement of some applications (e.g.: frequency regulation, voltage regulation, fluctuation smoothing and micro-grid control).

- **Transient fault recording function**

The system can continuously record the signals before and after the fault. The recorded data files are stored in the shared directory of the operator workstation and can be used for fault analysis. The file format adopts the standard COMTRADE format.

2.1.3 Features

- It is developed based on high-performance high-reliability UAPC platform of NR and provides a friendly human-machine interface.
- Configured with charging and discharging limit function, so as to cooperate with BMS to ensure battery safety during charging and discharging.
- Provides complete protection function to ensure reliable and safe operation of the product, or shutdown safely and rapidly.

- The quality of the grid-connected power is far superior to the technical requirements of relevant standards.
- Adaptive to high altitude applications: <5000m (>3000m, derating).
- With CAN, RS-485, RJ45, FO (Fiber Optics) and other communication interfaces, easy for communication.

2.2 Appearance



Figure 2.2-1 Appearance of the MV skid

The MV skid mainly consists of a Ring Main Unit (RMU), a 33kV/0.69kV transformer, four 2500kW PCSs and a container.

2.3 Electrical Structure

The electrical structure of the MV skid is shown in the following figure. Bidirectional converter topology is adopted to realize bidirectional power flow for charging and discharging.

In this structure, when the energy storage unit is charging, the MV skid operates as a rectifier, which converts the system side AC to DC, and stores the energy into the energy storage unit. When the energy storage unit is discharging, the MV skid operates as an inverter, which converts the energy released by the energy storage unit from DC to AC, and feeds it back to the external system. The PCS (Power Conversion System) AC CB (circuit breaker) is capable of realizing the grid-connected control function. The output of the converter is connected to a step-up transformer, so as to match the voltage of the parallel AC network, and to isolate the energy storage system from the external system. The RMU (Ring Main Unit) is equipped with protection devices with measurement and control function, so as to ensure the normal operation of the MV skid reliably.

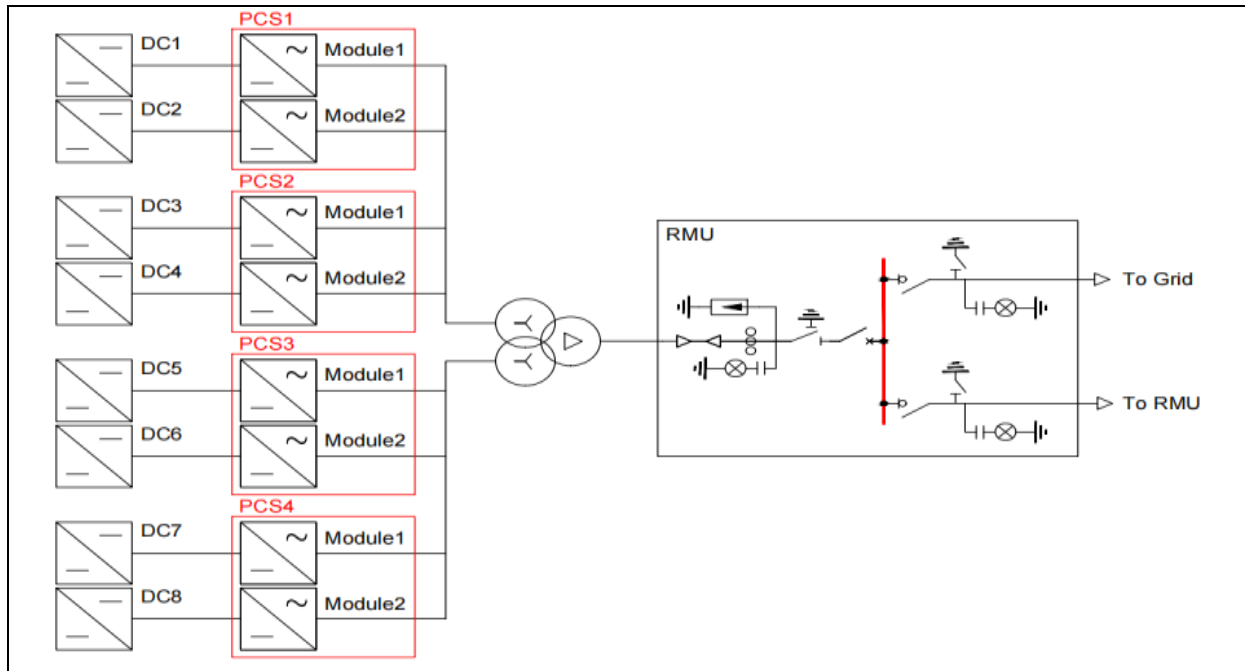


Figure 2.3-1 Main diagram of MV skid

The external 33kV bus is connected via RMU, converted to 690V AC voltage by transformer, and then connected to the AC side of the 4 PCSs. Meanwhile, the DC sides of the 4 PCSs are connected to corresponding battery stacks respectively.

2.4 Communication Topology

PCS-9567 PCS control unit (abbreviated as “**PCS-9567 control unit**” hereinafter) communicates with the BMS (Battery Management System) in real-time to obtain the status information of the batteries, and operates in charging or discharging mode according to the BMS monitoring data and control commands, achieving the purpose of energy transmission between the batteries and the AC grid. Meanwhile, the control unit can monitor the operation conditions of the whole power conversion system, ensure its reliable operation under overvoltage, overcurrent, BMS fault conditions and etc.

● Communication with EMS

PCS-9567 control unit and PCS-9611S can communicate with EMS via RS-485 or Ethernet, and various protocols are supported: IEC-60780-103, IEC-61850, Modbus, and so on.

PCS-9567 control unit supports fiber optic high-speed communication with the upper coordination controller and GOOSE protocol. It can cooperate with the upper coordination controller to form a high-speed control network and realize the fast control function.

● Communication with BMS

The control unit can communicate with the BMS system, monitor real-time status of the batteries and provide batteries alarm and fault protection, ensure safe operation of batteries. It supports CAN, Modbus TCP and Modbus RTU communication modes.

By default, this product communicates with BMS in accordance with Modbus TCP protocol, which has the characteristics of fast communication speed and high stability. If adopting Modbus RTU protocol, Modbus gateway (optional) is required to realize the conversion between Modbus TCP and Modbus RTU.

The Modbus protocol gateway includes an Ethernet port and a serial port (RS-232/RS-422/RS-485). It can automatically translate between Modbus TCP (Ethernet) protocol and Modbus ASCII/RTU (serial) protocol without additional programming or work.

- **Typical communication system**

The typical communication system of the MV skid is shown in the following figure. It realizes communication with the EMS and BMS via Ethernet.

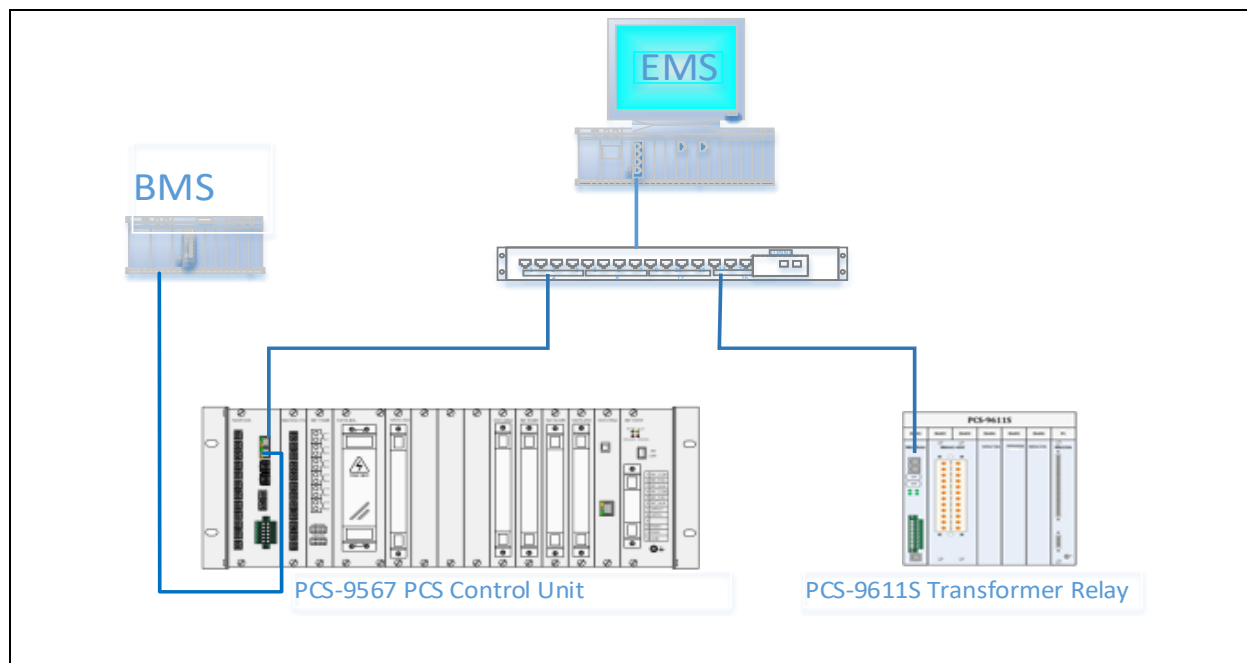


Figure 2.4-1 Typical network communication structure

2.5 Operation Modes

PCS-9567 control unit supports 2 operation modes: “**grid-following mode**” and “**grid-forming mode**”.

2.5.1 Grid-following Mode

In the “**grid-following mode**”, PCS-9567 control unit controls the charging and discharging of energy storage unit according to the following charging and discharging modes.

- **Constant DC voltage mode**

When this mode is enabled, the PCS will charge and discharge the energy storage unit according to the pre-configured constant DC voltage value.

- **Constant AC power mode**

When this mode is enabled, the PCS will charge and discharge the energy storage unit according to the pre-configured constant AC power value.

- **Automatic switching into constant voltage mode at the end of discharging or charging**

When this mode is enabled, the PCS will automatically switch to the constant DC voltage mode to charge and discharge the energy storage unit at the end of discharging or charging. It is necessary to set the limits for recognizing the end of charging and discharging according to the specific battery characteristics, as well as the corresponding voltage value after switching to constant DC voltage mode.

In the **“grid-following mode”**, the PCS will charge and discharge the energy storage unit according to the pre-configured internal control parameters. The charging and discharging modes can be switched freely. The relevant configuration (including the operation mode and parameters) can be modified on the local HMI or remote HMI.

The above operation modes need to be carried out under the premise of ensuring the safety of energy storage unit, and the PCS will continuously exchange information with the BMS to confirm the safe operating range of the energy storage unit. If the current operation state exceeds the safe operating range of the energy storage unit, the PCS will automatically switch the charging/discharging current or power to the safe area provided by the BMS.

2.5.2 Grid-forming Mode

When **“grid-forming mode”** is enabled, the PCS will simulate the operation characteristic of virtual synchronous generator: change the active power by changing the power angle, and change the reactive power by changing the voltage amplitude. Its overall characteristics of voltage source can realize the natural release of energy. Meanwhile, it has certain self-discipline ability to meet the needs of parallel operation of multiple voltage sources. In this mode, the external characteristics of the energy storage converter are similar to those of the conventional synchronous unit, it has the main functions of inertia support, frequency regulation and voltage regulation, and increasing the short circuit capacity of the conventional unit. It can solve or improve the problems of low inertia and insufficient support capacity of the current power grid, and enhance the support capacity of the regional power grid. This control technology can really play a supporting role of voltage source in the power grid.

The PCS in **“grid-forming mode”** can automatically operate in **“grid-connected state”** or **“off-grid state”**.

- **“Grid-connected state” to “Off-grid state”**

When the power grid is de-energized, or the PCC (Point of Common Coupling) switch is open, the PCS will switch to the **“grid-forming mode”** automatically or according to the grid-forming command and continue to provide the stored energy for the load.

- **“Off-grid state” to “Grid-connected state”**

When the power grid recovers to the **“grid-connected state”**, or switch to **“grid-connected state”** by schedule, the PCS will synchronize with the power grid automatically or according to the grid-connected command, and the PCC switch will be closed.

2.5.3 Switch Operation Mode

PCS can realize online switching between “**grid-following mode**” and “**grid-forming mode**” according to actual requirements without shutdown operation.

2.6 Operation States

PCS-9567 control unit can operate in these states: “**standby state**”, “**running state**”, “**fault state**”, “**emergency stop state**” and “**shutdown state**”.

2.6.1 Standby State

When the PCS operate in “**standby state**”, if local or remote operation state switching command is received, it can switch to corresponding state quickly.

2.6.2 Running State

In “**running state**”, the PCS can operate in “**grid-connected mode**” or in “**independent inverter mode**”, and it can realize bidirectional power flow in grid-connected running.

2.6.3 Fault State

If any fault is detected, the PCS will automatically stop running and enter the “**fault state**”, and the fault information will be displayed on the LCD of the control unit for the user to check. The system continuously monitors whether the fault is eliminated or not, if so, the system will switch to the “**shutdown state**” and wait for a startup command, if not, the system will remain the “**fault state**”.

2.6.4 Emergency Stop State

When pressing the “**EMERGENCY STOP**” button, the system will enter the “**emergency stop state**”. After pressing the “**EMERGENCY STOP**” button, switch the AC circuit breaker to the “**OFF**” position. When it is confirmed that the power conversion system is allowed to run again, switch the AC circuit breaker to the “**ON**” position.

2.6.5 Shutdown State

The shutdown state means the PCS has been stopped according to the normal stop process manually.

2.7 Protection Functions of PCS-9567 Control Unit

PCS-9567 PCS control unit detects the operation state of the PCS, energy storage unit and power grid in real-time. Once a fault or abnormality is detected, the control unit will issue the corresponding alarm signal or stop command.

- **AC overvoltage protection**

During the operation of converter, the default max permissible power grid AC voltage at the power grid interface is 110% of the rated voltage. The max permissible AC voltage can be configured by corresponding setting.

When the AC voltage is greater than corresponding setting, the converter will be stopped, and then disconnected from the power grid. Corresponding alarm signal will be issued and displayed on the LCD of PCS-9567 PCS control unit.

- **AC undervoltage protection**

During the operation of converter, the default min permissible power grid AC voltage at the power grid interface is 90% of the rated voltage. The min permissible AC voltage can be configured by corresponding setting.

When the AC voltage is lower than corresponding setting, the converter will be stopped, and then disconnected from the power grid. Corresponding alarm signal will be issued and displayed on the LCD of PCS-9567 PCS control unit.

- **AC overfrequency/underfrequency protection**

During the operation of converter, the default permissible range of the power grid frequency at the power grid interface is 45Hz~55Hz (when the rated frequency is 50Hz), or 55Hz~65Hz (when the rated frequency is 60Hz). The permissible range of the power grid voltage can be configured by corresponding setting.

When the power grid frequency exceeds corresponding setting range, the converter will be stopped. Corresponding alarm signal will be issued and displayed on the LCD of PCS-9567 PCS control unit.

- **AC overcurrent protection**

During the operation of converter, the default max permissible AC current output is 110% of the rated current. The max permissible AC current output can be configured by corresponding setting.

When the AC current output is greater than corresponding setting, the converter will be stopped, and then disconnected from the power grid. Corresponding alarm signal will be issued and displayed on the LCD of PCS-9567 PCS control unit.

- **Negative-sequence overcurrent protection**

During the operation of converter, when the AC side negative-sequence current is greater than corresponding setting, the converter will be stopped. Corresponding alarm signal will be issued and displayed on the LCD of PCS-9567 PCS control unit.

- **Negative-sequence overvoltage protection**

During the operation of converter, when the AC side negative-sequence voltage is greater than corresponding setting, the converter will be stopped. Corresponding alarm signal will be issued and displayed on the LCD of PCS-9567 PCS control unit.

- **Islanding detection**

The converter adopts dual islanding detection: active detection and passive detection. When islanding fault is detected, the energy storage unit will be disconnected from the power grid within 0.2~2s. Corresponding alarm signal will be issued and displayed on the LCD of PCS-

9567 PCS control unit.

- **DC overvoltage protection**

The default max permissible DC side voltage input is 1500V. The max permissible DC side voltage input can be configured by corresponding setting.

When the DC side voltage input is greater than corresponding setting, the converter will be stopped, the energy storage unit will be disconnected from the power grid within 0.2~1s. Corresponding alarm signal will be issued and displayed on the LCD of PCS-9567 PCS control unit.

- **DC neutral-point voltage unbalance protection**

When the difference between the positive and negative voltages of DC side is greater than corresponding setting, the converter will be stopped instantaneously, the energy storage unit will be disconnected from the power grid within 1s. Corresponding alarm signal will be issued and displayed on the LCD of PCS-9567 PCS control unit.

- **DC undervoltage protection**

During the operation of converter, when the DC side voltage input is lower than corresponding min permissible voltage setting, the converter will be stopped instantaneously. Corresponding alarm signal will be issued and displayed on the LCD of PCS-9567 PCS control unit.

- **Reverse connection protection of DC input power cable**

During the operation of converter, if the DC input power cable is reversely connected, the converter will be stopped instantaneously and the DC switch will be opened. Only when the connection is correct, the converter will be put into operation.

- **DC overcurrent protection**

During the operation of converter, when the DC side current is greater than corresponding max permissible setting, the converter will be stopped within 0.2~1s. Corresponding alarm signal will be issued and displayed on the LCD of PCS-9567 PCS control unit. This setting should be configured in coordination with the charging and discharging limitation current of the energy storage unit.

- **Insulation detection**

Before startup of the converter, the insulation condition of DC side is monitored in real time. If insulation abnormality is detected, the converter can NOT be started. Corresponding alarm signal will be issued and displayed on the LCD of PCS-9567 PCS control unit.

- **Resonance protection**

During the operation of converter, the three-phase voltages on the power grid side are monitored in real-time. When the harmonic component of any phase voltage on the power grid side exceeds the limit, or the number of zero-crossing points of any phase current exceeds 250 times within 1 second, or any phase capacitor current is greater than the overcurrent protection setting, the converter will be stopped. Corresponding alarm signal will be issued and

displayed on the LCD of PCS-9567 PCS control unit.

- **Driving protection**

During the operation of converter, the state of IGBT (Insulated Gate Bipolar Transistor) module is monitored in real time. If the driving fault of IGBT is detected, the converter will be stopped instantaneously. Corresponding alarm signal will be issued and displayed on the LCD of PCS-9567 PCS control unit.

- **VT (VoltageTransformer) supervision**

During the operation of converter, the voltage deviation between the two sides of the AC circuit breaker is monitored in real time. If any abnormal voltage measurement is detected, the converter will be stopped instantaneously. Corresponding alarm signal will be issued and displayed on the LCD of PCS-9567 PCS control unit.

- **Auxiliary power supply protection**

During the operation of converter, the state of PCS-9567-PWA auxiliary DC power supply is monitored in real time. If any auxiliary DC power fault is detected, the converter will be stopped instantaneously. Corresponding alarm signal will be issued and displayed on the LCD of PCS-9567 PCS control unit.

- **Over temperature protection**

During the operation of converter, the temperature of power modules is monitored in real time. When the temperature exceeds the upper setting, the air cooling will be started automatically and the converter will operate according to the pre-configured power range. When the temperature is greater than the ultra upper setting, the converter will be stopped, and it can resume running when the temperature recovers to normal range.

- **Communication failure protection**

During the operation of converter, it monitors the communication state between PCS-9567 PCS control unit with BMS & EMS), and the heartbeat messages (which requires the support of EMS) in real time. When the communication is interrupted, the converter will be stopped. Corresponding alarm signal will be issued and displayed on the LCD of PCS-9567 PCS control unit.

- **External interlock protection**

The converter can connect with the external interlocking protection signal. When the external interlocking protection operates, the converter will be stopped. Corresponding alarm signal will be issued and displayed on the LCD of PCS-9567 PCS control unit.

- **External emergency shutdown protection**

External emergency stop protection signal can be connected to the converter. When the external emergency stop signal is “1”, the converter will be stopped. Meanwhile, the AC CB (Circuit Breaker) and DC switch will be tripped, and the corresponding alarm message will be issued.

2.8 Protection Functions of PCS-9611S

- Overcurrent protection

During the operation of the converter, the AC output current will not exceed the overcurrent protection setting. When short circuit occurs in the power grid, and the AC output current may exceed the overcurrent protection setting. The RMU (Ring Main Unit) circuit breaker will be tripped by overcurrent protection, thus disconnected from the power grid. Corresponding alarm signal will be issued and displayed on the LCD of PCS-9611S.

- Transformer over-temperature protection

When the transformer is running normally, the temperatures of transformer winding/oil will not exceed the over-temperature protection setting. In case of transformer fault, or the ambient temperature is too high, the temperatures of transformer winding/oil may exceed the over-temperature protection setting. The RMU circuit breaker will be tripped by over-temperature protection, thus disconnected from the power grid. Corresponding alarm signal will be issued and displayed on the LCD of PCS-9611S.

- Transformer room door open protection

During normal operation of the system, the transformer room door and high-voltage room door are forbidden to be opened for operational safety. Once PCS-9611S detects door opening during operation, the RMU circuit breaker will be tripped by transformer room door open protection, thus disconnected from the power grid. Corresponding alarm signal will be issued and displayed on the LCD of PCS-9611S.

- Transformer Buchholz trip protection

Oil-immersed transformers will not generate gas during normal operation. If some faults (e.g.: short circuit, discharging and etc.) occur inside the transformer, the transformer oil may generate gas to trigger Buchholz trip protection. The RMU circuit breaker will be tripped by transformer Buchholz trip protection, thus disconnected from the power grid. Corresponding alarm signal will be issued and displayed on the LCD of PCS-9611S.

- Pressure relief protection

Pressure relief protection is a kind of safety protection. If some faults (e.g.: short circuit, discharging and etc.) occur inside the transformer, a large amount of gas will be generated to increase the pressure inside the transformer, so as to trigger the transformer pressure relief protection. The RMU circuit breaker will be tripped by pressure relief protection, thus disconnected from the power grid. Corresponding alarm signal will be issued and displayed on the LCD of PCS-9611S.



3 Commissioning

3.1 Inspection before Power-on

- Appearance inspection

All equipment of the this product is in good appearance, without damage, deformation or corrosion.

- Primary wiring inspection

AC and DC primary cable connection is firm and reliable, the phase sequence and positive/negative poles are wired correctly, and the cable terminal production process is qualified.

- Secondary wiring inspection

Check the external secondary wiring (emergency stop signal) is correct, firm and reliable.

The secondary control power wiring of the prefabricated substation is accurate and the phase sequence is correct.

- Communication wiring

Each communication port is accurately plugged and connected, and the appearance of the optical fiber is intact.

- Equipment internal inspection

(1) There is no missing tool inside the equipment. No tools are left inside the equipment.

(2) All sealing covers and filters of the equipment have been restored.

(3) Cabinet outlet hole should be blocked.

(4) The interior of the equipment is clean, free of debris, condensation and sand.

CAUTION

The intrusion of moisture and sandstorms is highly likely to damage the equipment! To ensure the normal use of various functions of the equipment, please comply with the following:

Do not open the cabinet door when the air humidity is above 95%;

Avoid opening cabinet doors to install, maintain, or inspect equipment in rainy, damp, or windy weather conditions.

3.2 Power on the Secondary System

3.2.1 Power on the Secondary System of PCS

During the commissioning period of some project sites, the conditions for primary system power on are not met. Therefore, temporary power supply need to be used on site for commissioning. When connecting the temporary commissioning power supply, the following aspects should be noted:

- (1) The temporary commissioning power supply on site is connected to the AC 220V.
- (2) To connect the temporary power supply, please select the first PCS that needs to be commissioned. The wires on the AC input terminal of the auxiliary power supply should be removed (please take photos of the terminal before removal for easy restoration later), and only the temporary commissioning power supply (L, N and ground wire) should be connected. For the commissioning of other PCSs, power can be temporarily taken from this terminal. After the commissioning of all PCSs is completed, please restore the terminal of the first PCS. Please refer to the following figure:

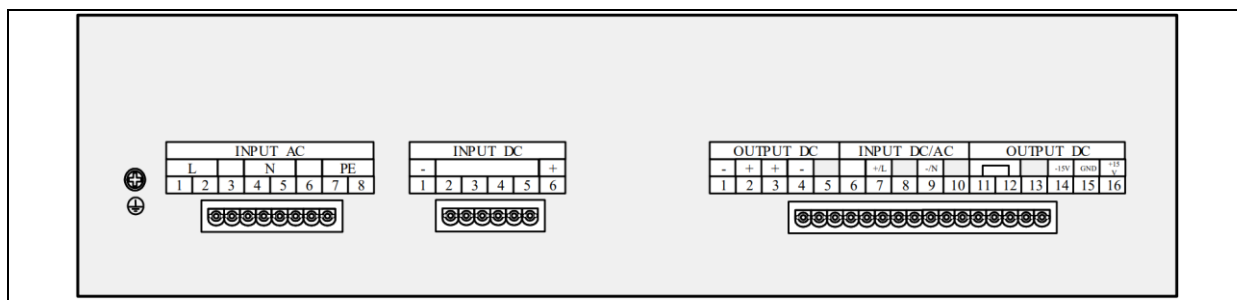


Figure 3.2-1 Commissioning power supply wiring diagram

- (3) Before powering on, make sure that the AC circuit breaker and DC switch are open, and the secondary MCB is in open.
- (4) Power on the secondary power supply.
- (5) Close the MCB 1K to energize the PCS control circuit, power on the PCS-9567 control unit.

CAUTION

Do NOT connect the secondary transformer to the commissioning power supply circuit, so as to prevent reverse power supply.

3.2.2 Prefabricated Substation Control Circuit Power Supply

- (1) Connect the external commissioning power line to the prefabricated substation control circuit terminal block according to the drawing.
- (2) Open all MCBs in the distribution box.
- (3) Check that the phase sequence of power lines is accurate.
- (4) Power on the control circuit.

- (5) Close the control circuit MCB, power on PCS-9611S, and now the control circuit of the prefabricated substation is energized.

3.3 Check Device Program Version

3.3.1 Check Program Version of PCS-9567 Control Unit

- (1) Connect the debugging network port (marked in the rectangle in the following figure) on the NR1233A module of PCS-9567 control unit by the PCS-PC software, and the IP of the debugging network port is 100.100.100.100.

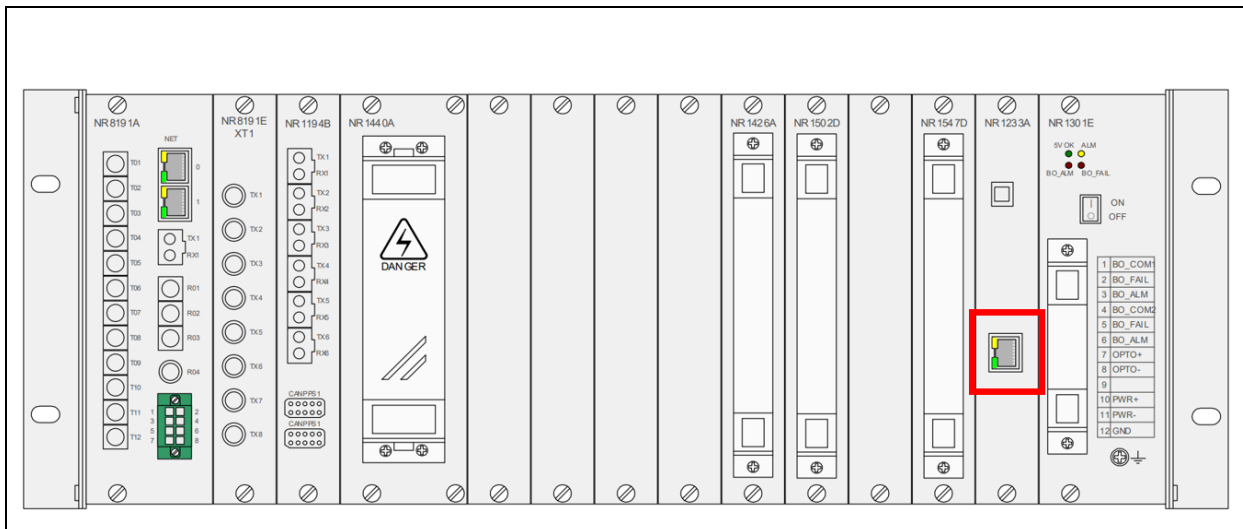


Figure 3.3-1

- (2) Check the device program, verify the onsite program version with our engineer. If a program upgrade is required, please contact our engineer to upgrade the onsite program.

3.3.2 Check Program Version of PCS-9611S



Figure 3.3-2 Front panel of PCS-9611S

Description of the front panel of PCS-9611S:

- (1) Local LCD screen for device status and information display
- (2) Easy-to-use keypad aiding simple navigation and set-point adjustment
- (3) Push buttons for opening/closing operation and switching of local/remote control mode
- (4) Push button for user login/logout of authority management
- (5) Push buttons with configurable labels for user-programmable functions
- (6) Front RJ45 multiplex port for debugging
- (7) Push button for target reset of all signals
- (8) Two LED indicators for device running status
- (9) Up to “1”8 (6U, 1/2 × 19”) programmable target LEDs with user-configurable labels

Procedure to check program version of PCS-9611S:

- (1) Check the onsite program version through the LCD display of this device. You can also connect the front RJ45 multiplex port for debugging (marked with “6” in the above figure) by the PCS-Studio software, and the IP of the network port is 100.100.100.100.
- (2) Verify the onsite program version with our engineer. If a program upgrade is required, please contact our engineer to upgrade the onsite program.

3.4 Confirm Settings

Export the settings of PCS-9567 control unit by PCS-PC software, and export the settings of PCS-9611S by PCS-Studio software, send them to our engineers by E-mail for check, and then import it to the device after check and confirmation.

3.5 Binary Inputs/Outputs Test

3.5.1 Binary Inputs/Outputs Test for PCS

- (1) Remotely open/close the DC switch by the PCS-PC software, so as to test the opening/closing functions.
- (2) Short the binary output terminals according to the drawing, so as to test the opening/closing functions of the AC circuit breaker.



DC switches can only be opened/closed through the device isolator links, remote control and panel selector switch: the open/close buttons of the DC

switch body cannot be operated directly. (There is no pre-charging logic for the DC switch body to open and close.)

⚠ DANGER

The AC circuit breaker of the outdoor cabinet, which adopts an electrically operated structure, must ensure that the external AC primary circuit is not energized if the opening/closing functions of the AC circuit breaker are tested by shorting the terminals.

3.5.2 Binary Inputs/Outputs Test for PCS-9611S

Table 3.5-1 Binary input and binary output list of PCS-9611S

No.	Binary Input Name	Binary Output Name
1	BI_01	BO_01
2	BI_02	BO_02
3	BI_03	BO_03
4	BI_04	BO_04
5	BI_05	BO_05
6	BI_06	BO_06
7	BI_07	BO_07
8	BI_08	BO_08
9	BI_09	BO_09
10	BI_10	BO_10
11	BI_11	BO_11
12	BI_12	BO_12
13	BI_13	BO_13
14	BI_14	
15	BI_15	
16	BI_16	
17	BI_17	
18	BI_18	
19	BI_19	
20	BI_20	
21	BI_21	
22	BI_22	
23	BI_23	
24	BI_24	
25	BI_25	
26	BI_26	
27	BI_27	
28	BI_28	
29	BI_29	
30	BI_30	

The binary input/output name can be modified according to onsite requirements.

⚠ DANGER

All binary inputs/outputs can only be tested when the main circuit is de-energized, i.e.: the test is forbidden when the main circuit is energized.

3.6 Transformer Test

Test the transformer in accordance with relevant local standards, the standards vary in different regions, conventional transformer test items are listed in the following table.

Table 3.6-1 Transformer test item list

No.	Test item	Description
1	Insulation resistance test	Measure the insulation resistance of the HVS (high-voltage side) winding, LVS (low-voltage side) winding, as well as the winding-to-ground insulation resistance by using an insulation resistance tester (e.g.: megohmmeter).
2	Winding DC resistance test	Measure the DC resistance of HVS/LVS windings with a DC resistance tester.
3	Ratio test	Measure the ratio of HVS/LVS windings with a ratio tester.
4	Polarity or group test	Use polarity tester or ratio tester to check the polarity or winding group of the transformer.
5	Oil quality test (oil-immersed transformer)	Analyze transformer oil to detect the electrical strength, moisture content, acid value, dielectric loss factor and etc.
6	Voltage withstand test	Carry out frequency voltage test or induction voltage test for HVS/LVS windings respectively.
7	Protection device check test	Check whether Buchholz relay, pressure relief valve, thermometer and other protective devices are intact.

3.7 Communication Test

3.7.1 PCS and BMS Communication Test

PCS requires to get the voltage, current, maximum charging/discharging power, SOC and other important data from BMS, so it needs to establish communication with BMS. The communication between PCS and BMS involves various communication types, including serial Modbus, network Modbus, CAN and GOOSE communication. Please select the appropriate communication method according to different communication configurations of the project.

- (1) Get the communication protocol of BMS.
- (2) Perform PCS communication configuration according to the protocol.
- (3) Set the logic link setting [Link_BMS_Ctrl] to “1”.

- (4) Check whether the alarm signal **"Alm_Comm_BMS"** is **"0"**, which means that the communication is normal.
- (5) Perform communication test with BMS to ensure that the data of each communication point are accurate.

3.7.2 PCS and PMS Communication Test

Communication between PCS and PMS can realize the unified and coordinated control of all PCSs on site.

PCS and PMS communication test are listed as below:

- (1) Set the settings [ActiveGrp_PCS] and [SeqNum_PCS] of the PCS. [ActiveGrp_PCS] represents the setting group No. of the PMS control in which the PCS is located (there are 4 control areas in the PMS, and each area can control 32 PCSs). [SeqNum_PCS] represents the control sequence number of the PCS (the PMS can control up to **"1"**28 PCSs, so [SeqNum_PCS] can only be set in the range of **"1-128"**).
- (2) Set the setting [Link_PMS_Ctrl] of the PCS to **"1"**.
- (3) Check whether the alarm signal **"Alm_Comm_PMS"** is **"0"**, which means that the communication is normal.

3.7.3 PCS and PCS Carrier Synchronous Communication Test

When multiple PCSs are connected to the same AC bus or DC bus, all PCSs need to be synchronized with carrier signals.

- (1) Determine the leader device and follower device for carrier communication according to the connection method of carrier fibers.
- (2) Through the PCS-PC software, set the leader device setting [En_Host_SynPulse] of to **"1"**, and set the leader device setting [En_SynPulsePhase] to **"0"**.
- (3) In all the follower devices, set the setting [En_Host_SynPulse] to **"0"**, and set the setting [En_SynPulsePhase] to **"1"**.
- (4) Access the setting list of the PCS, set **"SynPulseProt.Link"** to **"1"**.
- (5) Check whether the signal **"PulseSynErrProt.Op"** is **"0"**, which means that the carrier communication is normal.

3.7.4 PCS, PCS-9611S and EMS Communication Test

PCS-9611S and PCS support various protocol (e.g.: Modbus, IEC 60870-5-103 and IEC 61850) for upward communication with EMS.

Please select an appropriate communication method according to the onsite requirements.

3.8 Open-loop Test

Purpose:

After this product arrive on site, it is necessary to carry out the “**2-step open-loop test**” on the PCS, so as to ensure that the equipment is in normal condition before the grid-connection test.

Test steps:

Before the grid-connection test, perform the “**2-step open-loop test**”. Firstly, connect the device via the PCS-PC software. Set the logic links [Link_ZeroP_To_Standby], [27DC_Link_Trip_DC-CB] and [Link_BMS_Ctrl] to “0”, set the logic link [Link_Opn_AC-Contactor] to “1”.

(1) Zero-voltage open-loop test

Check and confirm that there is no alarm signal in the PCS, set [Link_OpenLoopTest] and [Link_En_PWM] to “1”, now the device automatically triggers the protection DFR (Disturbance & Fault Recording) to start. Check and confirm that the PCS is not driving the protection signal, and then set [Link_En_PWM] to “0”, set [Link_OpenLoopTest] to “0”.

(2) Full-voltage open-loop test (disconnect the battery to prevent the fault range from extending to the battery side)

Power on the battery side, close the PCS DC switch, after the PCS DC bus charging is completed, de-energize the battery side, and then the DC voltage of the PCS side will drop slowly at a rate of about 1V/s. Set the logic links [Link_OpenLoopTest] and [Link_En_PWM] to “1”, now the pulse will be deblocked, now the device automatically triggers the protection DFR to start. Then set [Link_En_PWM] to “0”, and set [Link_OpenLoopTest] to “0”, open the PCS DC switch, check the waveform, the voltage waveform of the converter side is a voltage waveform with a sinusoidal tendency, the 3-phase inverting voltage exceeds corresponding 3-phase current by about 90 degrees. Waveforms of the 3-phase current is reversed with the 3-phase filtering capacitor current waveforms, and the PCS has no protection driving signal.

After the full-voltage open loop is finished, it is necessary to restore the setting. Set [Link_ZeroP_To_Standby], [27DC_Link_Trip_DC-CB], [Link_BMS_Ctrl] to “1”; set [Link_Opn_AC-Contactor] to “0”.

3.9 Auxiliary Control Equipment Test

Perform power on test for each auxiliary control device (e.g.: lighting, outlets, temperature controlled heaters, fans, UPS and etc.), so as to ensure that the equipment is working properly.

3.10 Power on Power Supply Formally

After the primary/secondary formal power supply conditions are met, remove the previous temporary commissioning power supply wires, restore the original wiring, and carry out the

operation steps in the following sections.

3.10.1 Power on Secondary Power Supply

The auxiliary control power supply of the MV skid adopts external power supply, and send the secondary power supply of this project to the MV skid, and ensure that all auxiliary control devices in the MV skid can work normally, and there is no alarm signal.

3.10.2 Power on Main Circuit DC Power Supply

- (1) Open the earthing switch of the incoming cabinet in RMU firstly, then close the load switch.
- (2) Open the earthing switch of the outgoing cabinet in RMU firstly, then close the circuit breaker. Now the transformer is energized, and the lower side of the AC CB of PCS is energized.
- (3) Close the MCB in the PCS internal control circuit, as well as the device power supply, so as to power on the PCS control device.
- (4) After the device is powered on, use PCS-PC software to connect to the PCS-9567 control unit, so as to view the low voltage AC power grid voltage. Meanwhile, trigger DFR, and confirm the amplitude and the phase sequence of AC power grid voltage are correct.

WARNING

When the lower side of the DC switch of PCS is energized, it is strictly prohibited to manually close the AC circuit breaker. Manual closing may cause personal injury or equipment damage (the AC circuit breaker of this product adopts automatic startup logic, and manual operation is prohibited).

3.10.3 Power on Main Circuit DC Power Supply

Power on power supply for the DC battery side, check the DC battery side voltages in the PCS measurement list, and confirm the battery voltage values and polarities are normal.

3.11 Grid-connection Test

- (1) Remotely close the DC switch. Check the DC battery voltage and the DC bus voltage are the same.
- (2) Set the setting [Opt_Mode_P_AC_Ctrl] of PCS to “1”, and set [P_AC_Manual_Set] to 10% of rated active power. Set [Link_BMS_Ctrl] to “1”. Check and confirm that there is no alarm signal.
- (3) Set [Link_Start_Conv] to “1” by the commissioning software to start the PCS, or rotate the selector switch to “**START**” to start the PCS as shown in **Figure 4.1-4**.
- (4) After successful start, check and confirm that the PCS AC/DC active power values have been configured to the pre-configured active power. Check and confirm the PCS module

temperature is normal, the operation sound is normal, there is neither odor nor alarm signal. Check and confirm the transformer oil temperature is normal, the electrical parameters are normal, and there is no abnormal sound and odor. After normal operation, increase the active power to 30%, 50%, 70%, 100% till reaching the rated power. If the equipment arrives at the site and is stored for more than three months, it must be operated at 25% rated power for 24 hours without any problems before gradually increasing the power for debugging and operation.

- (5) If there is an extremely low temperature operating environment below -25°C on site, direct operation of on-site equipment (including PCS, switchgear, etc.) at extremely low temperatures may affect product performance, and it is necessary to first power on and heat for 30 minutes before starting operation.

WARNING

It is strictly prohibited to manually operate the AC circuit breaker of PCS body. Manual closing may cause personal injury or equipment damage (the AC circuit breaker of this product adopts automatic startup logic, and manual operation is prohibited)!

WARNING

It is strictly prohibited to operate the DC switch body mechanism to close the DC switch. The DC switch can only be closed by remote control from PCS-PC software or EMS!

3.12 Off-grid Test (Optional)

Off-grid test is an optional test according to onsite requirement. Before the off-grid test, please contact our engineers to confirm whether the off-grid test conditions are available, and whether relevant settings should be adjusted.

- (1) Before off-grid test, make sure that the high-voltage AC bus connected to the MV skid is not energized, and there is no load connected to the AC bus, as well as the bus is disconnected from the upper level grid connection point.
- (2) The circuit breakers of the RMU incoming and outgoing cabinets are closed.
- (3) Use PCS-PC software to set the logic link setting [Link_Island_Ctrl] of PCS to “1”.
- (4) Power on the battery side DC power supply, remotely close the DC switch of PCS.
- (5) Check and confirm there is no alarm signal in this device.
- (6) Start PCS.
- (7) After successful startup, check and confirm whether the AC voltage is normal, whether the operation sound and module temperature of the PCS are normal. Trigger DFR manually,

and confirm the waveform is normal.

- (8) After normal operation, the load can be gradually increased, but the load can NOT exceed the rated capacity of the PCS.

CAUTION

It is forbidden to put into the transformer after PCS off-grid operation, because the excitation inrush current is too big, which will lead to PCS overcurrent tripping. If there is actual requirement to put into the transformer, please put into the transformer before the transformer blackstart.

3.13 Charging and Discharging Test

In order to test the battery performance, the protection function of BMS and the operation capability of PCS, it is necessary to perform charging and discharging test for the battery after completing the onsite single unit commissioning. The specific test requirements can refer to the corresponding requirements of the battery manufacturer or the owner.

During the test, please pay attention to the following details:

- (1) Whether the temperature of PCS module is normal.
- (2) Whether the PCS fans can start and stop normally.
- (3) Whether PCS operation sound is normal, and whether there is any abnormal sound or odor.
- (4) Whether the PCS can execute the maximum charging/discharging power (maximum charging/discharging current) issued by the BMS.
- (5) Whether the PCS can execute the fully charging/discharging signal of BMS (usually standby after the fully charging/discharging).
- (6) Whether the PCS can execute the emergency stop signal from the BMS
- (7) Whether the PCS can perform the corresponding standby/stop operation after receiving alarm/fault signals from BMS.

CAUTION

When the main circuit of the equipment is live, it is forbidden to touch the AC/DC side protective barrier. Touching it may cause personal injury or equipment damage.

3.14 Whole System Commissioning

3.14.1 Commissioning for PCS & PMS

If PMS is integrated in this project, commissioning for PCS & upper PMS is required. Details are listed as below:

- (1) The communication between PCS and PMS is normal, and there is no abnormal alarm signal. The upper PMS can receive important parameters such as active power, operation state and SOC of PCS.
- (2) PMS can effectively control the start/stop and active power of PCS.
- (3) PCS can meet the power regulation rate required by PMS.

3.14.2 Commissioning for MV Skid & EMS

If EMS is integrated in this project, commissioning for this product (MV skid) & upper PMS is required. Details are listed as below:

- (1) The communications between EMS and PCS-9611S & PCS are normal, and there is no abnormal alarm signal.
- (2) The binary inputs and measurements of PCS and PCS-9611S received by EMS are normal.
- (3) EMS can effectively remote start/stop the PCS, and can open/close the DC switch remotely.
- (4) EMS can effectively open/close the circuit breaker of the incoming/outgoing cabinet in RMU.
- (5) EMS can effectively remote control PCS operation mode, as well as active power of remote regulation.

3.15 Start-Stop Procedure

3.15.1 Start Procedure

- (1) Open the earthing switch of the incoming cabinet in RMU firstly, then close the load switch.
- (2) Open the earthing switch of the outgoing cabinet in RMU firstly, then close the circuit breaker.
- (3) Close the MCB 1K in the PCS internal control circuit.
- (4) Power on the battery side, confirm the lower side of the DC switch of PCS is energized.
- (5) Remotely close the DC switch of PCS.
- (6) Check the PCS operation mode and the pre-configured active power value, confirm there is no alarm signal in this device.
- (7) Remotely start the PCS.

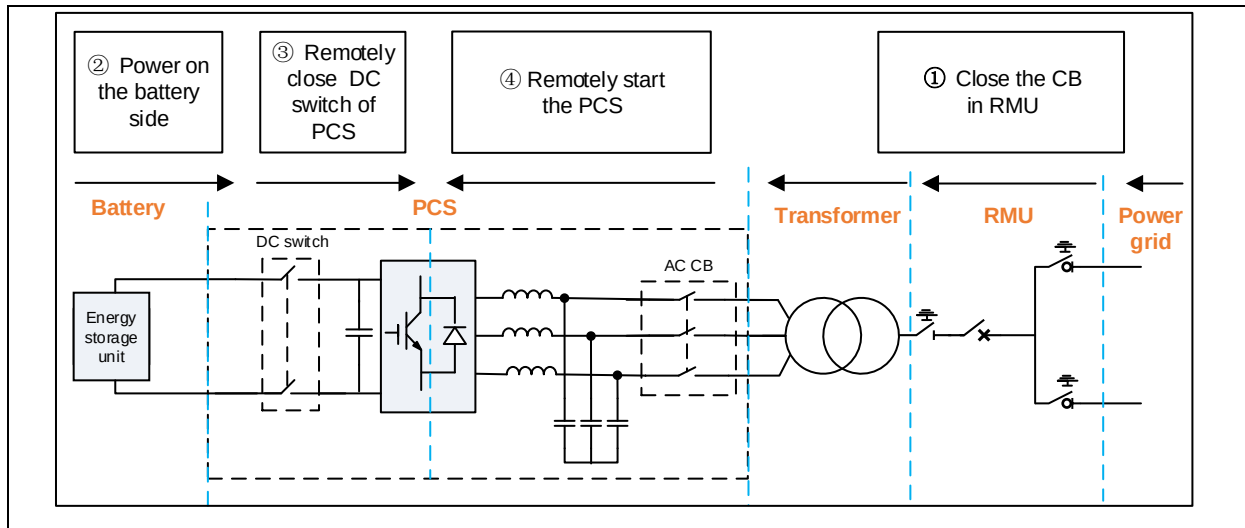


Figure 3.15-1 Flow chart of start procedure

3.15.2 Stop Procedure

- Normal stop procedure:
Remotely stop the PCS.
- Long-term stop or maintenance stop procedure:
 - (1) Remotely stop the PCS.
 - (2) Remotely open the DC switch of PCS, check the DC bus voltage of PCS, confirm that the voltage decrease to 0.
 - (3) Power off the battery side.
 - (4) Open the secondary control circuit power supply MCB 1K.
 - (5) Open the circuit breaker of the outgoing cabinet in RMU firstly, then close the earthing switch.
 - (6) For maintenance stop procedure, please take safety measures as required.

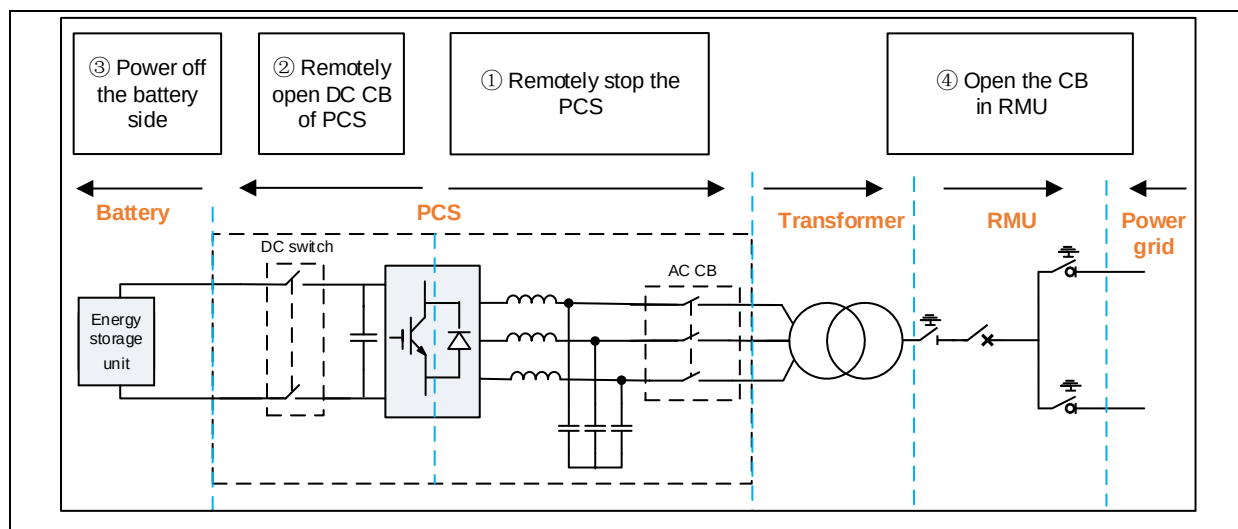


Figure 3.15-2 Flow chart of stop procedure



4 HMI Operation

4.1 Overview

The local HMI of PCS is located at the front of the cabinet, including “**STOP**” and “**RUN**” LED indicator, “**EMERGENCY STOP**” button, “**START/STOP**” selector switch. Please refer to the following figure for details.



Figure 4.1-1 Layout

- **LED indicators**

The LED indicators are used to indicate the operation state of the PCS.

Table 4.1-1 LED indicators

No.	Name	Color	Description
1	STOP	Green	PCS is out of service
2	RUN	Red	PCS is running

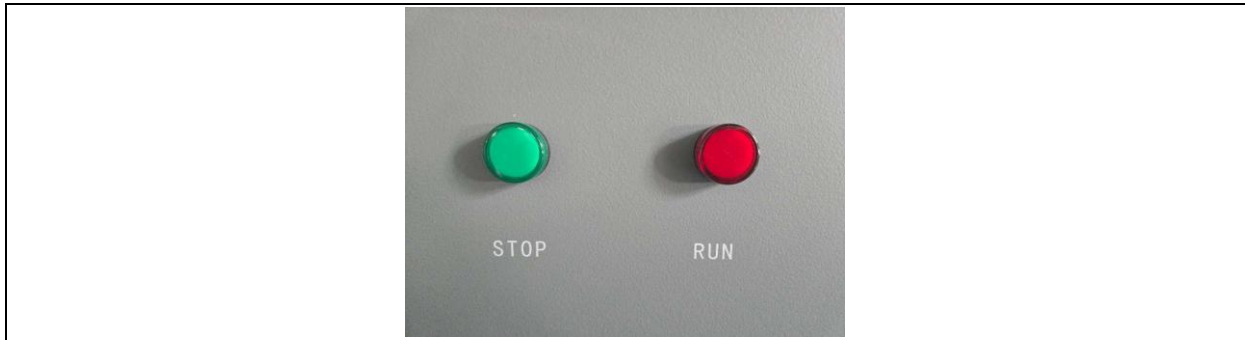


Figure 4.1-2 LED indicators

- **“EMERGENCY STOP”** button

Press down this button, the DC switch will be disconnected, the AC circuit breaker will be tripped, and the PCS will be out of service. This button is mainly used to immediately disconnect the PCS from the power grid in case of system fault. After the fault is cleared and before the system restart, please pay attention to lift the button up. Besides, in case of normal operation of the unit, please follow the procedure for normal shutdown operation. Do NOT press the emergency shutdown button at will.



Figure 4.1-3 “EMERGENCY STOP” button

- **“START/STOP”** selector switch

Rotate the selector switch to **“STOP”** to stop the PCS at once.

Rotate the selector switch to **“START”** to start the PCS.



Figure 4.1-4 “STOP/START” selector switch

4.2 Power on the Device

Before powering on this product, check and confirm that the power supply is wired correctly, both AC circuit breaker and DC switch are open. Toggle the button on the power supply board of the device, and the device will be powered on.

INFORMATION

Under normal circumstances, the start-up time of the device shall not exceed 30s. If the "STOP" indicator on the panel of PCS is not lit on after this time, it means that there is device startup abnormality, try to power off the device and restart it. If the fault still exists, please contact the manufacturer for resolution.

4.3 Instructions for Use

After connecting the device with the PCS-PC software, the following menu can be displayed in the status bar on the left side.

4.3.1 Measurements

This menu is used to display real-time measured values of this device. Please check the detailed values of this menu before putting this device into service.

4.3.2 Status

This menu is used to display status information, binary input signals and self-supervision alarm signals. Please check the values of the signals in this menu before putting this device into service.

Table 4.3-1 Description of submenu “Status”

No.	Submenu Name	Description
1	Operation State	Display operation state information of this device.
2	Contact_Inputs	Display the binary input information of this device.
3	Superv_Events	Display the self-supervision alarm states of this device.

4.3.3 Records

Various records of this device are displayed in this menu. Please check and record these records first after device operation.

Table 4.3-2 Description of submenu “Records”

No.	Submenu Name	Description
1	Disturb Records	Display protection operation of this device.
2	Superv Events	Display supervision events of this device.
3	I/O Events	Display binary input/output events of this device.
4	Device Logs	Display device logs of this device.
5	Control Logs	Display remote control logs of this device.
6	Clear Records	Clear all records.

4.3.4 Settings

This menu is used to check and modify the parameters and settings of this device.

Table 4.3-3 Description of submenu “Settings”

No.	Submenu Name	Description
1	Comm Settings	Check or modify the communication settings.
2	System Settings	Check or modify the system parameters.
3	Logic Links	Check or modify the logic links settings.
4	Protection Settings	Check or modify the protection settings.
5	BMS Settings	Check or modify the BMS settings.

Appendix A Acronyms and Abbreviations

AC	Alternating Current
BCU	Bay Control Unit
BI	Binary Input
BMS	Battery Management System
BO	Binary Output
CAN	Controller Area Network
CB	Circuit Breaker
CT	Current Transformer
DC	Direct Current
EMI	Electromagnetic Interference
EMS	Energy Management System
FO	Fiber Optics
GFM	Gird-Forming Mode
HMI	Human Machine Interface
HV	High Voltage
HVS	High-Voltage Side
IEC	International Electrotechnical Commission
IGBT	Insulated Gate Bipolar Transistor
IMD	Insulation Monitoring Device
LCD	Liquid Crystal Display
LED	Light Emitting Diode
LV	Low Voltage
LVS	Low-Voltage Side
MCB	Miniature Circuit Breaker



MV	Medium Voltage
PCB	Printed Circuit Board
PCC	Point of Common Coupling
PCS	Power Conversion System
PMS	Power Management System
PWM	Pulse-Width Modulation
RMU	Ring Main Unit
SOC	State of Charge
SOH	State of Health
SWG	Switchgear
THD	Total Harmonic Distortion
UPS	Uninterruptible Power Supply
VT	Voltage Transformer